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Re-evaluating low-molecular-weight heparin use in recurrent missed miscarriage: current evidence and future directions

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ABSTRACT

Background: Recurrent missed miscarriage (RMM) is heterogenous and emotionally burdensome. Low-molecular-weight heparin (LMWH) plus low-dose aspirin improves live-birth rates in women with obstetric antiphospholipid syndrome (APS), but the value of the use of LMWH in APS-negative RMM with and without confirmed coagulation disorders remains controversial, and optimal dosing frequency is uncertain.

Methods: We performed a narrative synthesis of randomised controlled trials (notably ALIFE and ALIFE2), Cochrane systematic reviews, pharmacokinetic studies, and national/international guidelines (RCOG, ESHRE). Emphasis was placed on live-birth outcomes, trial design limitations (including inclusion of chromosomally abnormal losses), pharmacokinetics relevant to once-daily (OD) versus twice-daily (BD) regimens, and safety/tolerability data.

Results: Robust evidence supports LMWH plus low-dose aspirin in improving live-birth rates in women with APS. In APS-negative populations, randomised controlled trials and meta-analyses show no reproducible live-birth benefit; interpretation is limited by clinical heterogeneity, underpowering and failure to exclude chromosomally abnormal losses, which account for up to half of early miscarriages. Notably, these trials almost exclusively used once-daily LMWH. Pharmacokinetic data indicate that twice-daily dosing yields more stable anti-Xa levels than once-daily dosing; however, this is a pharmacokinetic rationale only, and no adequately powered trial has tested whether it improves clinical outcomes.

Conclusions: Routine LMWH for APS-negative RMM is not supported outside clinical trials. Key knowledge gaps include the effect of pre-randomisation genetic stratification (euploid-only cohorts), directly comparing OD vs BD LMWH regimens, and incorporation of patient-centred and health-economic outcomes. Well-designed, adequately powered trials with genetic and phenotypic enrichment and embedded pharmacokinetic sub-studies are required to resolve clinical equipoise.

PLAIN LANGUAGE SUMMARY

Miscarriage is common, and some women sadly experience it again and again. This is very distressing, and doctors cannot always explain why it happens.

One possible cause is a problem with blood clotting, which may affect how the placenta (the organ that nourishes the baby) develops. Because of this, blood-thinning treatments have been tried to help prevent miscarriage. One such treatment is a daily injection of a blood thinner, often given together with a low dose of aspirin.

This review looked at how well these blood-thinning injections work. The evidence shows that they clearly help one specific group: women who have a known immune condition that makes the blood more likely to clot. For these women, treatment improves the chance of a successful pregnancy. For women who do not have this condition, however, research so far has not shown that the injections improve the chances of having a baby. Many studies are hard to interpret because they included miscarriages caused by genetic differences in the pregnancy, which no treatment can prevent.

There is also debate about how often the injection should be given each day. Giving it twice a day keeps the medicine at a steadier level in the body, but no good-quality study has shown that this leads to more babies being born. In summary, these injections should not be used routinely for women without the immune clotting condition, except within research studies. Future research needs to choose patients more carefully and to consider their experience and wellbeing.

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Introduction

Recurrent miscarriage (RM), variably defined as two or more or three or more consecutive pregnancy losses affects a small but important proportion of couples and causes substantial psychological and healthcare burden (Royal College of Obstetricians and Gynaecologists 2023). For clarity, in this review ‘recurrent miscarriage’ (RM) refers to two or more consecutive pregnancy losses, in line with current guidance, while ‘recurrent missed miscarriage’ (RMM) refers specifically to the subset of recurrent losses identified as missed (silent) miscarriages – that is, early foetal or embryonic demise detected on ultrasound in the absence of bleeding or symptomatic expulsion. As missed losses are more likely to reflect placental or coagulation-related mechanisms than losses presenting with bleeding, RMM is the population of principal interest when considering anticoagulant therapy. Aetiology is heterogeneous, recognised causes include chromosomal abnormalities, uterine structural anomalies, endocrine dysfunction, immunological disorders (notably antiphospholipid syndrome (APS)) and inherited or acquired thrombophilias. Chromosomal abnormalities (primarily embryonic aneuploidy) account for a large share of early losses; estimates cluster around 40–50% of sporadic first-trimester miscarriages (de Jong *et al.* 2011). Inclusion of chromosomal losses dilutes the effect of interventions targeting modifiable causes.

LMWH, usually combined with low-dose aspirin is supported by randomised evidence in women with obstetric APS, where anticoagulation is thought to reduce placental thrombotic events and improve live-birth rates (Ziakas *et al.* 2010, Empson *et al.* 2005). While APS increases miscarriage risk via thrombosis, other unrecognised prothrombotic states may contribute to unexplained RMM. Published studies as well as treatment centres stop short of investigating the whole total clotting pathway and therefore some of these patients will be labelled as RMM due to unknown causes. LMWH empirical use in women without APS remains contentious as large pragmatic randomised controlled trials (such as the ALIFE trial) and systematic reviews (notably the Cochrane review of aspirin and/or heparin for unexplained recurrent miscarriage) have failed to demonstrate a consistent live-birth benefit in unexplained RM, but interpretations are complicated by heterogeneity, small trials, and variable exclusion of chromosomal abnormalities (Kaandorp *et al.* 2010, De Jong *et al.* 2014).

A clinically important yet understudied question is whether LMWH dosing frequency affects outcomes. Twice a day (BD) dosing produces more consistent anti-Xa exposure than once daily (OD) from a pharmacokinetic perspective, but no sufficiently powered obstetric RCTs have investigated the efficacy of the BD dosage. Current obstetric guidance (e.g. RCOG Green-top) generally favours OD prophylactic dosing on pragmatic grounds (adherence, fewer injections, bleeding risk) rather than because of direct obstetric efficacy trials comparing dosing frequency (RCOG 2023). Given the continued empirical use of LMWH in some settings, a rigorous re-evaluation of dosing in genetically and clinically stratified cohorts is needed.

The specific contribution of this review is to draw together two strands that are seldom considered jointly. The under-examined question of LMWH dosing frequency (OD versus BD) and the confounding effect of chromosomally abnormal (aneuploid) losses and to use them to define the design features of a future trial capable of resolving the current clinical equipoise. We argue that persistently ‘neutral’ results in APS-negative populations may, in part, reflect both unselected (aneuploid-diluted) cohorts and potentially sub-optimal dosing, rather than proving the absence of any treatable subgroup.

Evidence sources and study selection

This narrative review synthesises randomised trials, systematic reviews, meta-analyses, and professional guidelines relevant to LMWH use in recurrent miscarriage and APS, emphasising studies reporting live birth and those addressing dosing regimen or pharmacokinetics. Although this is a narrative review and was not conducted as a systematic review, we searched PubMed/MEDLINE and the Cochrane Library (from inception to August 2025) using combinations of the terms ‘recurrent miscarriage’, ‘recurrent pregnancy loss’, ‘low-molecular-weight heparin’, ‘heparin’, ‘aspirin’, ‘antiphospholipid syndrome’, ‘thrombophilia’ and ‘pharmacokinetics’. We prioritised randomised controlled trials, Cochrane and other systematic reviews, pharmacokinetic studies, and national and international guidelines (RCOG, ESHRE) reporting live birth or dosing/pharmacokinetic data. Reference lists of key articles were hand-searched. Studies were selected by both authors on the basis of relevance to the review questions; no formal quality scoring or

meta-analytic pooling was undertaken. Major sources include randomised trials (ALIFE), Cochrane systematic reviews (aspirin/heparin for unexplained RM), the ALIFE2 trial results, and national/international guidelines (RCOG; ESHRE). Trial design limitations and safety data were examined to frame research priorities. A structured comparison of the principal trials and systematic reviews is provided in [Table 1](#).

LMWH in antiphospholipid syndrome (APS)

A substantial and consistent evidence base supports the use of antithrombotic therapy, typically LMWH combined with low-dose aspirin, in pregnant women with obstetric APS. Randomised trials, cohort series and subsequent meta-analyses demonstrate improved live-birth rates and reduced foetal loss when anticoagulation is instituted in women meeting APS criteria (Empson *et al.* 2005, Ziakas *et al.* 2010). The mechanistic rationale is strong: antiphospholipid antibodies interact with trophoblasts and the maternal endothelium, promote complement activation and encourage placental micro-thrombosis and infarction, all of which can compromise implantation and placental function. These antibody-driven effects reduce trophoblast invasiveness and placental perfusion, trigger local inflammation and complement deposition, and increase tissue-factor mediated coagulation within the intervillous space, a constellation that plausibly links antiphospholipid seropositivity to recurrent pregnancy loss and provides a clear biological target for antithrombotic therapy (Tong *et al.* 2015). Anticoagulation mitigates the thrombotic component of this pathology and thus improves pregnancy outcome in appropriately selected patients (Ziakas *et al.* 2010).

Clinically, this evidence underpins guideline recommendations to treat confirmed APS in pregnancy with combined aspirin and LMWH from diagnosis (or conception) through the majority of gestation, with tailoring of dose and duration according to maternal risk factors and bleeding history (RCOG 2023). Diagnosis requires persistent antiphospholipid antibodies on two occasions ≥ 12 weeks apart using validated assays, interpreted within appropriate clinical context to avoid misclassification (Devreese *et al.* 2020). Multidisciplinary input (obstetrics/foetal-medicine, haematology, rheumatology and specialised laboratory support) further optimises management and helps replicate trial-level outcomes in routine care (Empson *et al.* 2005).

LMWH for APS-negative or unexplained recurrent miscarriage

For women without a confirmed diagnosis of APS as per above criteria, current evidence from RCTs and pooled analyses does not demonstrate a clear improvement in live-birth rates with LMWH with or without aspirin. One of the largest and most methodologically robust RCTs in this field, the ALIFE trial, investigated the use of aspirin plus nadroparin compared with aspirin alone or placebo in women with unexplained RM and found no statistically significant benefit for the combination therapy (Kaandorp *et al.* 2010). However, it should be noted that this study (and most others) used OD dose of LMWH and the elimination half-life of nadroparin is only 3.7 hours raising the question of whether these patients were adequately covered by thrombo-prophylactics for the whole 24-hour period. Similarly, a Cochrane systematic review synthesising the available RCT evidence concluded that there is insufficient support to recommend routine use of aspirin and/or heparin for unexplained RM (De Jong *et al.* 2014). The review

Table 1. Summarises the key pharmacokinetic parameters of common LMWHs MW – molecular weight and SC – subcutaneous.

LMWH	Mean SC (Da)	Peak plasma anti-Xa (h post-dose, SC)	Anti-Xa:IIa ratio	Bioavailability	SC half-life
Enoxaparin	~ 4500	3–5	~3.6-3.8:1	~98% (Sanderink <i>et al.</i> 2002)	4.5 (single) (Sanderink <i>et al.</i> 2002)
Dalteparin	~6000	3–5	~2:1	~87% (Samama and Gerotziakas 2000)	3-5 (Samama and Gerotziakas 2000)
Nadroparin	~4300	4-6 (Samama and Gerotziakas 2000)	~ 3.8:1	~98% (Samama and Gerotziakas 2000)	~8-10 (Samama and Gerotziakas 2000)
Certoparin	~4200	2–3	~4-5:1	~96%	~2

Da, daltons; h, hours; LMWH, low-molecular-weight heparin; MW, molecular weight; SC, subcutaneous.

highlighted key limitations in the existing literature, including small sample sizes, heterogeneity in study populations and interventions, and variable definitions of RM across trials.

It is important to distinguish this pharmacokinetic plausibility from demonstrated clinical efficacy: that OD dosing may leave sub-therapeutic troughs is a hypothesis generated by pharmacokinetic data, not evidence that BD dosing would improve live birth. No trial has tested this directly (see Dosing frequency, below).

Chromosomal abnormalities and other non-modifiable causes of loss

Many trials did not exclude embryonic chromosomal abnormalities, a non-modifiable cause responsible for approximately 40–50% of early miscarriages. Inclusion of such losses dilutes treatment effects, leading to underestimation of benefit for interventions targeting modifiable mechanisms (De Jong *et al.* 2014, de Jong *et al.* 2011).

Pharmacology and elimination kinetics of LMWH

LMWH are depolymerised fragments of unfractionated heparin that retain predominantly anti-factor Xa activity with relatively less anti-IIa action, yielding improved subcutaneous bioavailability and a more predictable dose–response than unfractionated heparin (UFH). Common agents in obstetric practice (enoxaparin, dalteparin, tinzaparin) have modestly different elimination profiles: enoxaparin's elimination half-life is roughly 4–5 hours, dalteparin ~3–5 hours, and tinzaparin ~3–4 hours in non-pregnant adults, although assay and product differences produce some variability between reports (Novak *et al.* 2023, Imbalzano *et al.* 2024, Amerali and Politou 2022). LMWH differ in molecular weight and anti-Xa activity, which in turn influence their pharmacokinetics. Enoxaparin (~4.5 kDa) and nadroparin (~4.3 kDa) both demonstrate anti-Xa: anti-IIa ratios of around 3.5–3.9:1 and exhibit very high subcutaneous bioavailability (~90–98%) (Gray *et al.* 2008, Garcia *et al.* 2012). In contrast, dalteparin (~6.0 kDa) has a lower anti-Xa: anti-IIa ratio and slightly reduced bioavailability (~86–87%) (Samama and Gerotziafas 2000). These variations impact absorption and elimination: enoxaparin reaches peak anti-Xa activity at 3–5 hours post-dose with an elimination half-life of ~4.5–7 hours (Hirsh and Raschke, 2004), nadroparin peaks at ~4–6 hours with a longer half-life of ~8–10 hours, and dalteparin displays a half-life of ~3–5 hours (Mismetti *et al.* 2001, Sanderink *et al.* 2002). Certoparin, another LMWH, has a relatively short half-life of ~2 hours.

Furthermore, pregnancy-associated physiological changes, notably expanded plasma volume, increased glomerular filtration and altered protein binding typically reduce circulating drug exposure and can shorten apparent half-life, meaning fixed once-daily prophylactic regimens may produce lower trough anti-Xa activity in pregnancy than in non-pregnant adults (Eke 2022, Coppola *et al.* 2022) (Rodger *et al.* 2014). As LMWH is predominantly renally cleared, maternal weight and renal function materially influence exposure and support consideration of weight-adjusted dosing or targeted anti-Xa monitoring in extremes of body habitus or renal impairment (Sikes *et al.*, 2023); however, routine monitoring in pregnancy remains controversial and is practised selectively (Wu *et al.*, 2020). These pharmacological considerations provide the biological rationale for testing twice-daily schedules or dose adjustment strategies in obstetric trials rather than assuming one-size-fits-all dosing. They do not, however, by themselves establish that such strategies improve pregnancy outcomes; that remains a hypothesis requiring direct trial evidence.

Dosing frequency: OD versus BD

These kinetic differences imply that BD regimens would achieve more stable anti-Xa profiles with reduced peak–trough variability compared to OD dosing. For example, comparative data indicate that enoxaparin produces higher trough anti-Xa levels than nadroparin when administered at equivalent mg/kg dosing, suggesting more consistent anticoagulant coverage (Greer and Nelson-Piercy 2005). In theory, steadier exposure through BD dosing could reduce transient sub-therapeutic troughs and the risk of placental microthrombosis. It must be emphasised, however, that this is a pharmacokinetic rationale rather than demonstrated clinical benefit: stable anti-Xa exposure is a surrogate, and whether it

translates into higher live-birth rates is unknown. There is currently no randomised controlled trial evidence directly comparing OD and BD prophylactic LMWH regimens for miscarriage prevention. Major guidelines, such as those of the RCOG (2023), recommend OD prophylaxis in pregnancy primarily for pragmatic reasons, including adherence, convenience and reduced bleeding risk, rather than proven clinical superiority. Thus, the OD versus BD dosing question remains unresolved and represents an important target for future trials that incorporate pharmacokinetic sub-studies alongside obstetric outcomes.

Beyond frequency, the choice of dose itself (fixed prophylactic versus weight-adjusted or intermediate dosing) is a further variable that has not been standardised across trials and should be prespecified in future studies. Taken together, the OD-versus-BD question (and the related dose-selection question) remains an area of genuine clinical equipoise and warrants targeted trials integrating pharmacokinetic sub-studies (e.g. anti-Xa profiling) with clinical endpoints, to determine whether more stable anticoagulation translates into improved pregnancy outcomes. Until such evidence exists, the hypothesised advantage of BD dosing should be presented as a rationale for research rather than a basis for changing practice (Table 2).

Safety and acceptability

LMWH is generally safe in pregnancy especially that they do not cross the placental barrier to the foetus, but it is not without adverse effects on the mother. Injection-site reactions are common and can reduce adherence; clinically significant bleeding is uncommon but must be considered, as should the rare risk of clinically significant heparin-induced thrombocytopenia (HIT) even with LMWH (Ziakas *et al.* 2010). The ALIFE trial reported higher rates of local adverse effects in LMWH recipients without concomitant efficacy benefit in APS-negative women, a finding that emphasises the ethical and economic implications of exposing large cohorts to daily injections when benefit is unproven (Kaandorp *et al.* 2010). Patient-centred outcomes, including injection acceptability, impact on mental health, cost and quality of life have been under-reported in the RM literature but are crucial when considering prophylactic LMWH in a largely APL-negative population (Bai *et al.* 2021). Notably, because the total daily dose is unchanged, splitting it into two 12-hourly half-doses lowers the peak anti-Xa level and may therefore reduce bleeding risk should injury or an unplanned intervention occur – a potential safety argument for BD dosing that, like the efficacy argument, remains to be tested formally. Practical clinical decisions therefore require not only efficacy data but also robust safety profiles and measures of treatment burden; such outcomes should be prespecified secondary endpoints in future trials.

Table 2. Summary of key random.

Study (design)	Population and key inclusion criteria	Intervention vs comparator	Live-birth outcome	Key finding and limitations
Kaandorp <i>et al.</i> 2010 – ALIFE (RCT, n=364)	Unexplained RM (≥ 2 losses); APS excluded; chromosomal losses not excluded	Aspirin + nadroparin (OD) vs aspirin vs placebo	No significant difference in live birth between groups	No benefit in unselected population; OD dosing only; aneuploid losses not excluded
Quenby <i>et al.</i> 2023 – ALIFE2 (RCT, n=326)	RM with confirmed inherited thrombophilia	LMWH (OD) vs standard care	No live-birth benefit from LMWH	Negative even in a thrombophilia-enriched cohort; OD dosing; open-label
De Jong <i>et al.</i> 2014 (Cochrane systematic review; 9 RCTs, n $\approx$ 1,228)	Unexplained RM $\pm$ inherited thrombophilia	Aspirin and/or heparin vs placebo/no treatment	No overall improvement in live birth	Marked heterogeneity; variable RM definitions; little genetic/phenotypic selection
Empson <i>et al.</i> 2005 (Cochrane systematic review)	Recurrent loss with antiphospholipid antibodies / lupus anticoagulant	Heparin + aspirin vs aspirin alone	Reduced pregnancy loss / improved live birth with heparin + aspirin	Benefit confined to APS; limited number of trials
Ziakas <i>et al.</i> 2010 (systematic review and meta-analysis)	APS with recurrent pregnancy loss	Heparin (UFH or LMWH) + aspirin vs aspirin	Higher live-birth rate with combined therapy (chiefly UFH)	Supports anticoagulation in APS; heterogeneity between UFH and LMWH

ised controlled trials and systematic reviews of LMWH ($\pm$ aspirin) in recurrent miscarriage and antiphospholipid syndrome.

APS, antiphospholipid syndrome; LMWH, low-molecular-weight heparin; OD, once daily; RCT, randomised controlled trial; RM, recurrent miscarriage; UFH, unfractionated heparin.

Synthesis: clinical implications and research priorities

The clinical question of whether LMWH should be used, and at what dose and frequency, in women with RMM must be viewed through the prism of aetiology, trial evidence and pragmatic clinical considerations. Two clear and separate evidence streams emerge. First, in women with confirmed APS, anticoagulation with heparin (usually LMWH) plus low-dose aspirin is supported by RCTs and meta-analyses and is embedded in clinical guidance; the pathophysiology in APS (antibody-mediated placental thrombosis and trophoblast dysfunction) provides a mechanistic rationale for this benefit (Ziakas *et al.* 2010, Empson *et al.* 2005). Second, for women without confirmed APS – a heterogeneous group commonly presenting with unexplained RM, in whom the whole clotting pathway may not have been thoroughly investigated – the randomised evidence does not demonstrate a consistent live-birth advantage with LMWH, either alone or added to aspirin (Kaandorp *et al.* 2010, De Jong *et al.* 2014).

As discussed above, the repeated neutral results in APS-negative populations have several interacting explanations, of which dilution by non-modifiable (aneuploid) losses is the most important; in an unselected cohort, such losses attenuate any true treatment effect. Large, pooled evidence (nine RCTs, $n \approx 1,228$ in the Cochrane review) found no overall benefit from aspirin, LMWH, or their combination in unexplained recurrent miscarriage, but the included trials were heterogeneous in design, and many lacked careful genetic or phenotypic selection (De Jong *et al.* 2014).

By contrast, the evidence for anticoagulation in APS is more favourable, albeit still limited. In a Cochrane review of APS-related pregnancy loss, Hamulyák *et al.* (2020) found that the addition of heparin (unfractionated or low-molecular-weight) to low-dose aspirin may increase live-birth rates in women with persistent antiphospholipid antibodies and recurrent miscarriage; however, this apparent benefit was driven largely by a single large trial and the overall certainty of evidence was rated low. Importantly, no major maternal or foetal adverse events, including heparin-induced thrombocytopenia or congenital malformations were reported, a finding consistent with the established pharmacology that neither UFH nor LMWH cross the placenta (Hamulyák *et al.* 2020, Che Yaakob *et al.* 2010). Current recommendations therefore endorse heparin use in APS-positive patients (commonly combined with aspirin), while emphasising the limited strength of the supporting trial data (De Jong *et al.* 2014; Hamulyák *et al.* 2020).

To illustrate the impact of effect size and sample size on detectability, we undertook a transparent, hypothetical calculation using the Cochrane comparison of LMWH and aspirin versus no treatment (pooled $n=322$ for this comparison) as a pragmatic ‘large’ sample frame (De Jong *et al.* 2014). Because the pooled report does not publish a single pooled control event proportion useable for this exercise, we used a conservative baseline live-birth rate of 57% taken from the large ALIFE trial (Kaandorp *et al.* 2010) as a realistic comparator for non-APS RMM.

Assuming equal allocation ($n=161$ per arm) and a hypothetical doubling of treatment effect ($OR \approx 2.05$), the live-birth rate would increase to $\sim 73\%$ versus 57% in controls ($p \approx 0.0023$). This illustrates that, if a true effect of this magnitude existed in a biologically enriched subgroup, existing sample sizes might have been sufficient to detect significance. This calculation is purely illustrative and makes no claim about the real effect size; its purpose is to show that the problem may be one of patient selection rather than of sample size alone. However, dilution by non-modifiable losses and lack of phenotypic stratification may obscure clinically meaningful subgroup effects. The key practical message remains: dilution by non-modifiable losses and lack of prospective genetic/phenotypic enrichment can mask clinically meaningful subgroup effects; therefore, future trials should stratify or enrich by biologically plausible markers (for example euploid embryo selection, clear bleeding phenotype, or molecular placental signatures) and prespecify dosing comparisons to resolve whether a sizeable, clinically important benefit exists in a defined population.

Recurrent miscarriage is clinically heterogeneous, and this must inform trial design and therapy selection. For example, losses preceded by bleeding (or threatened miscarriage) may benefit from progesterone, whereas recurrent missed losses without bleeding suggest placental/coagulopathic causes. Broader causes, uterine anomalies, endocrine disorders, immunologic factors (including APS), inherited thrombophilias, and unexplained cases should be distinguished. Coagulation-targeted treatments cannot affect losses due to structural anomalies or aneuploidy, so trials that do not prospectively phenotype and

stratify participants risk diluting subgroup effects. Consequently, contemporary guidelines favour trial designs that select or stratify by biologically plausible subgroups (RCOG 2023, ESHRE 2023).

There is also inter-trial heterogeneity in the intervention itself, LMWH is not a single, uniform exposure. Different studies have used different LMWH agents, doses and durations, and nearly none have directly compared OD with BD prophylactic regimens. Pharmacokinetic data demonstrate that BD dosing produces smaller peak–trough anti-Xa variability compared with OD dosing, suggesting theoretically more continuous anticoagulant exposure (Patel *et al.* 2013). Whether that pharmacokinetic advantage translates into improved placental outcomes, for example, by reducing brief windows in which microthrombi may form remains untested in adequately powered obstetric trials.

The largest pragmatic trials further support restricting LMWH to APS. The ALIFE trial (aspirin and nadroparin vs aspirin vs placebo) in unexplained RM found no significant improvement in live birth with anticoagulation in an unselected population (Kaandorp *et al.* 2010). More recently, the ALIFE2 trial (women with recurrent loss and confirmed inherited thrombophilia) reported no live-birth benefit from LMWH compared with standard care (Quenby *et al.* 2023). Along with systematic reviews, these trials strengthen the conclusion that routine LMWH use outside of APS is not supported by high-quality evidence (De Jong *et al.* 2014).

Safety, tolerability, and patient burden are important in decision-making. LMWH is generally safe but has downsides: injection-site reactions are common, and although heparin-induced thrombocytopenia is rare with LMWH, bleeding complications can occur (Ziakas *et al.* 2010, Kaandorp *et al.* 2010). In ALIFE, women receiving LMWH reported more local adverse effects than controls, despite no live-birth benefit (Kaandorp *et al.* 2010), suggesting that daily injections impose psychological and economic costs. A recent patient survey also underscored the burden of daily injections on quality of life (Bai *et al.* 2021). These observations highlight the need to include patient-centred outcomes in future trials.

Current professional guidance has trended towards OD prophylactic dosing in obstetrics, largely on pragmatic grounds such as improved adherence and reduced injection burden, rather than on direct efficacy data in the RMM population (RCOG 2023). Pharmacokinetic investigations in pregnancy demonstrate that BD dosing produces more consistent anti-Xa levels, which could theoretically offer more stable anticoagulant coverage if intermittent placental microthrombi are clinically relevant (Patel *et al.* 2013). However, such studies were conducted in selected populations (e.g. women with APS, thrombophilia, or high BMI), limiting generalisability. Without randomised trials comparing OD versus BD in genetically and clinically stratified RM populations, clinical equipoise persists. Future head-to-head trials are therefore justified to determine whether BD dosing benefits any biologically defined subgroup or whether OD dosing is sufficient for most women.

Future trials should incorporate genetic and phenotypic stratification to yield actionable answers. Key design elements include:

- **Pre-randomisation genetic screening** (e.g. euploid embryo selection) or stratification to exclude expected non-viable conceptions.
- **Rigorous phenotyping** to distinguish APS, other thrombophilias, and unexplained RM.
- **Randomised comparisons of OD vs BD LMWH** regimens, using commonly used agents and weight-adjusted dosing as appropriate.
- **Embedded pharmacokinetic substudies** (e.g. serial anti-Xa levels) to correlate drug exposure with outcomes.
- **Adequate sample size and power** to detect clinically meaningful improvements after accounting for expected non-modifiable losses.
- **Inclusion of safety, quality-of-life, and economic endpoints** to ensure balanced, patient-centred recommendations.

Given these considerations, a pragmatic clinical stance is warranted. For women with confirmed obstetric APS, LMWH plus low-dose aspirin remains the recommended, evidence-based therapy (Ziakas *et al.* 2010, Empson *et al.* 2005). For APS-negative RM, clinicians should avoid routine LMWH outside of trials and should prioritise comprehensive evaluation (including foetal karyotyping when feasible) before empirical anticoagulation. If LMWH is considered in individual cases (for example, after specialist

consultation or patient request), the limited and uncertain evidence, potential adverse effects, and treatment burden must be discussed in shared decision-making.

The literature supports a focused and conservative approach: limit LMWH to confirmed indications (principally APS) and prioritise high-quality, phenotype-specific trials. Only through such targeted, methodologically rigorous studies (for example, testing OD vs BD dosing in genetically stratified RM populations) can we determine whether clinically meaningful benefits exist in defined subgroups. Such evidence will enable truly evidence-based, patient-centred recommendations on LMWH use in recurrent miscarriage.

Conclusion

In conclusion, RMM is a heterogeneous condition with significant emotional and healthcare consequences. Current high-quality evidence supports the use of LMWH plus low-dose aspirin for women with clotting disorders due to APS. In APS-negative recurrent miscarriage, however, randomised trials and systematic reviews do not show a consistent live-birth benefit, even when clotting parameters are not fully investigated. Key limitations include unselected trial populations, inclusion of chromosomal losses, and inconsistent LMWH regimens. Pharmacokinetic data suggest twice-daily dosing yields steadier anti-Xa exposure than once-daily dosing, but this remains a pharmacokinetic rationale only, as no adequately powered obstetric trial has compared these regimens for live-birth outcomes. Future research should prioritise adequately powered trials that carefully consider phenotype, with pre-randomisation genetic stratification (or euploid-only analyses), prespecified subgroup or enrichment designs, direct OD versus BD comparisons using weight-adjusted regimens, and embedded pharmacokinetic, safety, quality-of-life and health-economic endpoints. Such studies are needed to offer patients the best possible care at a profoundly difficult time.

Author contributions

CRedit: **Jaiden Townsend:** Methodology, Writing – original draft, Writing – review & editing; **Rami Attala:** Conceptualization, Supervision, Validation, Writing – review & editing.

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